CHANDRACHUD PATI

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RESEARCH INTERESTS

Quantum computer systems and architecture; quantum compilation and compiler diagnostics; noise-aware execution on NISQ hardware; quantum resource estimation, reliable quantum query processing; full-stack hardware–software co-design.

EDUCATION

National Institute of Technology Rourkela · Rourkela, India *2019–2024*
Integrated M.Sc. in Mathematics

Selected coursework: Quantum Computation, Quantum Mechanics, Probability and Statistics, Stochastic Processes, Numerical Analysis, Linear Programming, Graph Theory, and Number Theory.

PUBLICATIONS AND MANUSCRIPTS

[1] **Per-Phase Fidelity Attribution for Quantum Compilers using HBR Decomposition**
Chandrachud Pati and Yogesh Simmhan. *Accepted, IEEE Quantum Week (QCE) 2026. arXiv:2605.07876.*

[2] **Title withheld for double-blind review**
Chandrachud Pati et al. *Under review, 33rd IEEE International Conference on High Performance Computing, Data & Analytics (HiPC) 2026.*

[3] **Title withheld for double-blind review**
Chandrachud Pati and Yogesh Simmhan. *Under review, SFWM, co-located with SC 2026.*

RESEARCH EXPERIENCE

Indian Institute of Science (IISc) · Bengaluru, India *Feb 2026–Present*
Pre-doctoral Researcher, DREAM Lab · Advisor: Prof. Yogesh Simmhan

- **Compiler and resource assurance:** Designed HBR decomposition to attribute fidelity loss across compiler phases and SDKs; developing one-round shot provisioning that reuses completed-query statistics and preserved precision with half the submissions in an IBM hardware campaign.
- **Quantum/HPC runtime systems:** Under India's **National Quantum Mission** (NQM), developing resource aware execution policies and application-aware recovery for hybrid scientific workflows, connecting device level evidence to reliable runtime decisions.
- **Quantum data systems:** Developing an Apache Calcite admission and recovery layer that validates SQL-to-quantum plans before cost-based selection and answer release.

National Institute of Technology (NIT) Rourkela · Rourkela, India *Sep 2023–Jan 2024*
Research Project – Hybrid Classical–Quantum Machine Learning · Advisor: Prof. Snehashish Chakraverty

- Designed a hybrid neural network with two-qubit parameterized quantum circuits implemented in Qiskit and PennyLane as trainable Keras layers for highly imbalanced credit-card fraud detection.
- Achieved an AUC of 0.9433, about two percentage points above a matched classical baseline; observed that transpilation fidelity affected model accuracy, motivating later compiler-fidelity work.

INDUSTRY EXPERIENCE

American Express · Gurugram, India *Aug 2024–Jan 2026*
Engineer II, Next Generation Data Management

- Built an LLM-based data-architecture documentation system that reduced manual documentation effort by approximately 60% and contributed to an early promotion to Engineer II.
- Developed Python REST APIs for infrastructure provisioning and worked with enterprise-scale distributed data systems.

IDFC FIRST Bank · Mumbai, India *Feb 2024–Jun 2024*
Data Science Intern, New Age Special Projects

- Fine-tuned BERT and open-source language models for real-time transaction classification, reaching approximately 99% accuracy on an internal evaluation set.
- Built PySpark and AWS pipelines for ingesting, preprocessing, and analysing large-scale transaction records.

Tally Solutions · Bengaluru, India *May 2023–Jul 2023*
Engineering Intern, CTO's Office

- Designed Go chaincode and REST APIs for a Hyperledger Fabric platform supporting MSME credit workflows; containerized peers, orderers, certificate authorities, and application services.

HONORS AND AWARDS

2026	ACM SIGMOD/PODS India Fellowship – Quantum Data Management	Recipient
2025	American Express Early Careers Hackathon – Synthetic data generation with GANs	Winner
2023	IDFC FIRST Bank FAME 2.0 – EMI default prediction and collections modelling	Runner-up
2023	Flipkart GRID 5.0 – Diabetic-retinopathy detection using Quantum computing	Semi-finalist
2023	Unstop UG Leaders national competition	All-India Rank 4
2022	Tally CodeBrewers – Systems programming for file tagging	Runner-up

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TECHNICAL SKILLS

Quantum systems: Quantum compilation, compiler-phase attribution, backend-aware execution planning, resource estimation, shot allocation, circuit cutting, noise-aware benchmarking

Quantum software: Qiskit, PennyLane, TKET, Cirq; circuit transpilation, noisy simulation, compiled-circuit analysis

Systems engineering: Python, PySpark, SQL, Apache Calcite, REST APIs, distributed workflows, Git, Linux, AWS, Docker, Kubernetes

Machine learning: PyTorch, TensorFlow/Keras, graph neural networks, transformer and LLM fine-tuning, retrieval-augmented generation, GANs